Master Program Design

Oumph × Government Polytechnic, Bhubaneswar Centre of Excellence in Applied AI & Intelligent Systems

MoU dated 8 April 2026 · Initial Term 3 Years

Document 01 of 8 · Oumph internal + GPB partner use · v1.0

OOUMPH × GOVERNMENT POLYTECHNIC BHUBANESWAR

CENTRE OF EXCELLENCE — MASTER PROGRAM DESIGN

Program Title: Applied AI Workforce Engineering & Industry Integration Hub **Partners:** MyOoumph Networks Pvt. Ltd. (Industry Partner) × Government Polytechnic, Bhubaneswar (GPB / Institute) **MoU Execution Date:** 8 April 2026 **Initial Term:** Three (3) Years (8 April 2026 – 7 April 2029) **Prepared for:** Steering Committee review + CoE Academic Coordinator rollout **Document Owner:** Oumph CEO (Praveen Mishra) — approval gate

---

1. STRATEGIC CONTEXT

1.1 Why this program exists

The MoU creates a **Centre of Excellence (CoE)** whose job is *not* fundamental AI research but **applied AI workforce production**. Oumph uses this CoE as a talent-and-compute feeder into its 40-industry DAO roadmap (see §4); GPB uses it as an NEP-2020 / Skill-India / IndiaAI-Mission-aligned differentiator that converts polytechnic diploma students into employable Applied AI technicians, data specialists, and MLOps operators.

1.2 Oumph's Vision (for internal alignment — not all of this goes on the MoU board)

Oumph is building three interlocking layers:

1. **FLM 6.0** — Oumph's sovereign AI OS layer: model-agnostic, memory-native, multi-agent. The CoE's agentic builds plug into FLM 6.0 runtime so every student project ships on real Oumph infrastructure.
2. **40-Industry DAO Network** — one autonomous agentic-AI business ("DAO") per vertical. Each

DAO is a revenue-bearing node: it sells a service (lead scoring for real estate, crop-disease detection for agriculture, tender-watch for MSME, etc.). **The CoE's 40 student-teacher groups each adopt one DAO.**

3. **Brand Wars** — Oumph's internal KPI-driven competition between brand cells. The CoE becomes Oumph's "Odisha Cell" and competes against other regional cells on DAO ARR, deployments shipped, faculty certifications earned, and interns placed.

1.3 Agentic AI Solution Stack (what students actually build)

Every DAO in this program is an **agentic AI system** — meaning it is not a single model, but a pipeline:

```
[Data Ingestion] → [Annotation / Preprocessing] → [Fine-tuned or prompt-engineered model]
   → [Agent Orchestrator (FLM 6.0 runtime / LangGraph / CrewAI)]
   → [Tool / API layer] → [Human-in-the-loop approval node] → [Deployment endpoint]
   → [MLOps: monitoring, drift detection, retraining trigger]
```

Every graduate can build, deploy, and operate a pipeline of this shape. That is the employment-grade outcome.

---

2. PROGRAM STRUCTURE AT A GLANCE

Dimension	Specification
Cohort size	200 students/year (MoU minimum: 120)
Cohorts over MoU term	3 (Cohort-26, Cohort-27, Cohort-28)
Program length	52 weeks (24 weeks taught + 28 weeks capstone & deployment)
Group count	**40 student-teacher groups** per cohort
Group composition	5 students + 1 faculty lead + 1 Oumph mentor
DAO assignment	1 DAO per group (see §4 catalog)
Teaching hours	720 hrs/student (14 hrs/wk × 52 wks incl. labs)
Credits (indicative)	24 academic credits, recognised under AICTE IIC
Certification	GPB Diploma Add-on + Oumph Applied AI Engineer Certificate
Infrastructure	Tier-1 RTX GPUs (0–6 mo) → Tier-2 multi-GPU (6–18 mo) → Tier-3 MLOps cluster (18–36 mo)
Delivery mode	On-campus CoE + co-located Oumph ops unit + hybrid cloud

---

3. THE 40 GROUPS × 40 DAOs (COHORT-26, Y1 DEFAULT)

[redacted]
MoU Clause 4.4:

Track A — Applied AI in Agriculture, Fisheries, Food (5 DAOs)

Group	DAO	Core AI Problem
G01	**AgriDAO**	Crop-disease detection, yield prediction
G02	**FishDAO**	Aquaculture water-quality + species sorting (Chilika/Odisha context)
G03	**DairyDAO**	Cattle-health monitoring, milk-yield forecasting
G04	**FoodProcDAO**	Defect detection on processing lines (rice mills, fryums)
G05	**HortiDAO**	Horticulture pest ID, storage-loss optimisation

Track B — Applied AI in Engineering, Mining, Infra (6 DAOs)

Group	DAO	Core AI Problem
G06	**MiningDAO**	Ore-grade CV, haul-truck predictive maintenance (Odisha iron-ore corridor)
G07	**SteelDAO**	Defect detection on rolled-steel, furnace energy optimisation
G08	**CivilDAO**	Crack detection in bridges/roads (applied CV)
G09	**RealtyDAO**	Property verification, title-document OCR (Ooumph RealtyCheQ synergy)
G10	**ConstructDAO**	Site-safety PPE detection, progress monitoring
G11	**ArchDAO**	Auto-BoQ from architectural drawings

Track C — Applied AI in Energy, Utilities, Mobility (5 DAOs)

Group	DAO	Core AI Problem
G12	**PowerDAO**	Transformer-fault classification, grid load forecasting
G13	**SolarDAO**	Panel-defect detection from drone imagery
G14	**WaterDAO**	Leak detection, water-quality sensor fusion
G15	**WasteDAO**	Waste-segregation CV, route optimisation
G16	**MobilityDAO**	Auto-rickshaw ridesharing demand forecast, parking occupancy CV

Track D — Applied AI in Healthcare, Pharmacy, Wellness (5 DAOs)

Group	DAO	Core AI Problem
G17	**HealthDAO**	Medical-image triage (X-ray, ECG strip)
G18	**PharmaDAO**	Prescription OCR + interaction-warning agent

G19	**MentalDAO**	Odia/Hindi crisis-hotline triage NLP
G20	**FitDAO**	Pose-estimation for sports, movement-therapy coaching
G21	**DiagDAO**	Lab-report structuring from PDF scans

Track E — Applied AI in Commerce, Retail, MSME (5 DAOs)

Group	DAO	Core AI Problem
G22	**RetailDAO**	Shelf-analytics CV, SKU out-of-stock alerts
G23	**VibeDAO**	Ooumph VibeStore personalisation + review sentiment
G24	**LogiDAO**	Last-mile route AI, dispatch agent
G25	**MSMEDAO**	Tender-watch + RFP response agent
G26	**FinDAO**	KYC OCR, anomaly detection on credit ledgers

Track F — Applied AI in Governance, Safety, Public Service (5 DAOs)

Group	DAO	Core AI Problem
G27	**GovDAO**	Grievance-classification + routing (CM helpline)
G28	**SafetyDAO**	Dial-112 audio triage, incident prediction
G29	**TrafficDAO**	ANPR, traffic-signal optimisation
G30	**LandDAO**	Land-record NER + anomaly detection
G31	**LegalDAO**	Judgment summarisation + precedent retrieval (Odia/English)

Track G — Applied AI in Media, Creative, Tourism (5 DAOs)

Group	DAO	Core AI Problem
G32	**StoryDAO**	Ooumph Story-Matters comic/film script assistant
G33	**MerchDAO**	Ooumph Mugged-Moments custom-merch generative design
G34	**TourDAO**	Odisha-tourism itinerary agent (multilingual)
G35	**EduDAO**	Ooumph Gurukul tutor agent (Odia/Hindi/English)
G36	**HandiDAO**	Handicraft provenance + pricing agent for artisans

Track H — Applied AI in Telecom, Security, Edge (4 DAOs)

Group	DAO	Core AI Problem
G37	**CyberDAO**	Phishing-URL classifier, log-anomaly detection

G38	**TelecomDAO**	Churn prediction, call-drop root cause
G39	**EdgeDAO**	Edge-AI kit (Jetson/Coral) reference-design squad
G40	**DataDAO**	Annotation-ecosystem ops squad — supports all 39 other DAOs with labelled data

> **Note:** G40 (DataDAO) is the service-squad. It does not build a customer-facing DAO; it builds the annotation pipeline + quality-control agent that all other groups consume. This models Ooumph's real ops.

---

4. GROUP FORMATION — HOW THE 40 GROUPS ARE ASSEMBLED

4.1 Student selection (per group of 5)

Each group must have a mix of profiles so the group can actually ship a pipeline:

Role	Profile	Count per group
Data Lead	Strong at SQL, Python pandas, Excel	1
Model Lead	Strongest at PyTorch, comfortable with math	1
Frontend / UX Lead	Can build Streamlit/Gradio/React basic UI	1
Ops Lead	Linux, Docker, comfortable with command line	1
Domain Lead	Reads, writes Odia/Hindi well; knows the DAO's industry (e.g., mining-town student for MiningDAO)	1

Assignment algorithm (run by CoE Academic Coordinator in Week 0):

1. Students submit skill-self-assessment + top-3 DAO preferences.
2. Coordinator bucket-sorts by skill; greedy-matches to DAO preferences.
3. Domain Lead slot is filled from geography/family-background signal (a student from Keonjhar → MiningDAO; a student from Puri → TourDAO).

4.2 Teacher allocation

GPB faculty strength is typically ~60 across all departments. We need ≥40 faculty to lead groups. Strategy:

- **Core CoE Faculty (8):** Computer Science + Electronics depts → lead Tracks A–H anchor roles.
- **Domain Faculty (24):** Civil, Mechanical, Electrical, Pharmacy, Architecture, Mining Engg → lead domain-heavy DAOs in their specialty.
- **Rotating Faculty (8):** visiting + Applied Science depts → support cross-cutting DAOs (Gov, Retail, Education).

Each teacher attends 80 hours of **Teacher Training Material** (see doc `02_TEACHER_TRAINING_MATERIAL.md`) before leading a group.

4.3 Ooumph mentor assignment

Ooumph assigns 8 senior engineers as Track Leads (one per track A–H). Each Track Lead mentors 5 DAO groups. Mentorship is **2 hrs/week per group** via video + async on the CoE Slack.

---

5. WEEKLY TIME BUDGET (PER STUDENT)

Bucket	Hours/week	Annual
Core lectures (with faculty)	6	312
Hands-on lab (Ooumph-designed)	4	208
DAO squad work (own project)	3	156
Industry mentor hour	1	52
Total	**14**	**728**

On top, 4 hrs/week of self-study is expected (tracked via FLM learning platform).

---

6. 3-YEAR PHASED ROADMAP

Year 1 — Cohort-26 (Apr 2026 – Mar 2027)

- Infrastructure **Tier-1**: RTX-class GPUs, Edge-AI kits, annotation workstations.
- Faculty Development: 2 FDPs (Jul 2026, Jan 2027); NITTTR MLOps certification cohort.
- Student enrolment: 200. 40 groups, 40 DAOs.
- **MVP deployment expectation**: every group has a functioning Streamlit demo by Week 26 and a containerised deployment by Week 52.
- Industry revenue target: **5 DAOs reach paying pilot** (Ooumph absorbs revenue, shares with GPB per Annexure A).

Year 2 — Cohort-27 (Apr 2027 – Mar 2028)

- Infrastructure **Tier-2**: Multi-GPU A100/H100 servers; CV+IoT integration lab; high-speed networking.
- Cohort-26 alumni become **TAs** and interns — CoE becomes self-perpetuating.
- DAO catalog refresh: 3 DAOs may be swapped based on Year-1 learnings.
- Industry revenue target: **15 DAOs at paying pilot; 3 DAOs at recurring revenue.**
- Faculty: 30% of faculty hold at least one AICTE/NITTTR AI certification.

Year 3 — Cohort-28 (Apr 2028 – Mar 2029)

- Infrastructure **Tier-3**: MLOps cluster, containerized deployment, model registry, hybrid cloud.
- 40 DAOs operate as a portfolio; each has its own FLM 6.0 instance.
- Industry revenue target: **25+ DAOs revenue-positive**; CoE financially self-sustaining as agreed

under MoU §3.7.
- GPB becomes a **replicable model** (MoU §2.6) — Ooumph + Odisha govt can clone into 5 more polytechnics.

---

7. HYBRID REVENUE-SHARING MODEL (PROPOSED FOR ANNEXURE A)

The MoU (§10, §11) says revenue sharing is defined in annexures — so here is the proposed structure, which maps onto the 5 IP Categories from MoU §7.2.

7.1 Revenue streams covered

1. **DAO SaaS revenue** — monthly subscriptions to end-customers using an Ooumph-deployed DAO built during the CoE program.
2. **Consulting** / **custom-build revenue** — one-off projects sold via CoE brand.
3. **Paid certification fees** — outsiders paying to take CoE's certification.
4. **Industry affiliate membership fees** (MoU §11).
5. **MSME compute-access fees** for GPU hours.
6. **Government training contracts** routed via CoE.

7.2 The hybrid formula

Base split by IP Category:

IP Category (MoU §7.2)	Ooumph share	GPB share	Student/Faculty pool
A — Academic exploratory (IP with students)	0%	85%	15%
B — Industry problem statement	70%	20%	10%
C — Fully industry-funded	85%	10%	5%
D — Joint projects	50%	35%	15%
E — Public welfare (open-source)	30% (operating cost)	30% (academic credit)	40% (impact bonus for students)

Hybrid sliding adjustment (applied on GPB's share in Categories B, C, D):

- **Year 1:** Base share (as above) — infrastructure subsidy period.
- **Year 2:** GPB share +5% across B/C/D (infrastructure amortised).
- **Year 3+:** GPB share +10% across B/C/D (CoE self-sustaining).

Additional kickers for GPB on top of the base share:

- **GPU utilisation fee:** Ooumph pays GPB ₹X/GPU-hour when CoE GPUs are used for Ooumph's own co-located workloads (MoU §4.2).
- **Placement bonus:** ₹25,000 per Ooumph-placed student (on top of the student's internship stipend).

- **Faculty consulting fee:** Ooumph pays faculty ₹2,000/hr for consulting on non-CoE projects after Year 1.

Student/Faculty pool distribution:

- 60% of pool → the 5 students in the DAO squad (pro-rata by effort logged in FLM platform).
- 30% → faculty lead + Ooumph mentor of that DAO.
- 10% → a CoE fund (hackathons, equipment, prizes).

7.3 Settlement cadence

- Revenue recognised monthly; settlement quarterly.
- Statement format: CoE Finance Dashboard (auto-generated from Ooumph ERP → shared with GPB CoE Academic Coordinator + Accounts Officer).
- Audit: annual, by a mutually-appointed CA.

---

8. GOVERNANCE (per MoU §8)

8.1 Steering Committee (quarterly)

Seat	From Ooumph	From GPB
Chair (rotates)	CEO (Praveen Mishra)	Principal (Dr. Rajat Panigrahy)
KPI owner	VP-Tech & Strategy (Meetam Hyder)	CoE Academic Coordinator
Infra owner	Ayush (IT/Infra)	HOD Electronics
Curriculum owner	Avinash (AI Team)	HOD Computer Science

8.2 Executive Committee (weekly, operational)

Ooumph Track Leads × GPB Track Faculty. Resolves day-to-day issues.

8.3 KPI Dashboard (MoU §8.3 — the six KPIs)

KPI	Year 1 target	Year 2 target	Year 3 target
Students trained	200	200	200
Applied projects completed	40 (1/DAO)	60	80
GPU utilisation rate	≥ 45%	≥ 65%	≥ 80%
Internship→Employment conversion	≥ 20%	≥ 30%	≥ 40%
Faculty certifications earned	15	30	45
Subsidy funds leveraged	₹2 Cr	₹5 Cr	₹8 Cr

All KPIs are visible in a shared FLM dashboard accessible to both parties.

9. DATA, COMPLIANCE, RESPONSIBLE AI (per MoU §9)

The CoE follows **NITI Aayog Responsible-AI principles** verbatim. Specific commitments:

1. **No facial-recognition deployment** without written Steering-Committee approval (per MoU §9).
2. **Bias audits:** every DAO runs a bias-audit report in Week 40; shipping requires a passing audit.
3. **Data minimisation:** student projects use **synthetic or anonymised** datasets by default. Raw identifiable data is only used under Industry Problem Statements (Category B) with explicit data-sharing agreements.
4. **Model card:** every DAO ships with a model card (intended use, limits, dataset lineage, bias notes) — template in `07_RUBRICS_AND_ASSESSMENT.md`.
5. **30-day publication embargo:** any student paper/poster referencing CoE work is sent to Oomph 30 days pre-publication (MoU §7.5).

---

10. KEY DEPENDENCIES & FIRST-90-DAY CHECKLIST

#	Action	Owner	Deadline
1	Annexure A (revenue-sharing) drafted & signed	Oomph Legal (Mansi) + GPB Principal	30 Apr 2026
2	CoE Academic Coordinator nominated	GPB Principal	15 Apr 2026
3	Tier-1 hardware procurement PO	GPB + Oomph advisory	30 May 2026
4	Faculty cohort (40) identified and signed onto FDP	GPB HODs	15 May 2026
5	Student selection (200) + skill audit	GPB + Oomph BI Analyst	15 Jun 2026
6	CoE space handover with power/cooling	GPB Infra	15 Jun 2026
7	Oomph Track Leads (8) onboarded	Oomph HR (Shantanu)	30 Apr 2026
8	FLM 6.0 tenant spun up for GPB	Oomph Engineering (Sanaur)	15 May 2026
9	DAO catalog v1 signed off	Steering Committee	30 May 2026
10	Cohort-26 inauguration	Joint	01 Jul 2026

---

11. RISKS & MITIGATIONS

Risk	Likelihood	Impact	Mitigation

Hardware delivery delay	Medium	High	Ooumph provides hybrid-cloud burst capacity in Months 0-6
Faculty capability gap	Medium	High	80-hr FDP + Ooumph Track-Lead mentorship in parallel
Student attrition mid-year	Medium	Medium	Tiered milestones; groups can absorb 1 drop-out per 5
DAO fails to reach paid pilot	High	Medium	Revenue target is for 5/40 in Y1; rest is learning value
Data-privacy incident	Low	High	Data Governance Officer (Mansi, GPB Coordinator) pre-signs every dataset use
IP dispute	Low	High	IP category tagged at Week 0 for every DAO; written sign-off in FLM
Subsidy funds delayed	Medium	Medium	Ooumph bridges via Affiliate Membership revenue

---

12. DELIVERABLE INDEX (THIS PACKAGE)

#	File	Audience
00	`00_MASTER_PROGRAM_DESIGN.md` (this doc)	Steering Committee, executive
01	`01_SYLLABUS_AND_SCHEDULE.md`	CoE Academic Coordinator, curriculum reviewers
02	`02_TEACHER_TRAINING_MATERIAL.md`	40 GPB faculty leading groups
03	`03_STUDENT_TRAINING_MATERIAL.md`	200 cohort students
04	`04_ASSIGNMENT_BRIEFS_40_DAOs.md`	40 groups — their "industry problem brief"
05	`05_CODE_SNIPPETS_LIBRARY.md`	Students + faculty — copy/paste reference
06	`06_LAB_EXERCISES.md`	Week-by-week hands-on labs
07	`07_RUBRICS_AND_ASSESSMENT.md`	Grading + joint assessment panels

End of Master Program Design.